

UNDERSTANDING AND OPTIMISING MOBILITY



WHAT IS MOBILITY? FLEXIBILITY, ELASTICITY AND STIFFNESS

Mobility is a broader and more useful concept than others which we tend to use in the world of sport, such as flexibility or elasticity. Although mobility is affected to a certain degree by flexibility and elasticity, it cannot be defined solely by these terms. Mobility or range of motion (ROM) is, in fact, 'the amount of movement that exists in a joint and that is determined by the design of the joint' (Leal, Martínez and Sieso, 2012; Tortora and Derrickson, 2011; within Peláez Maza, 2015a).

There are several factors that affect the amount of movement in our joints:

- *The design of the joint itself*: this is determined by evolution and genetics so it's not something we can change. For example, the shape of our femoral neck bone will determine whether it's possible for us to do the splits.
- *Elasticity*: this is the ability of something (in this case, soft tissue) to return to its original state without deforming or breaking. Therefore, we can't be more elastic or less elastic: we either are or we aren't. There's no in between.
- *Muscle stiffness, tightness or tension*: this occurs when our muscles try to resist any form of deformation (or stretching). It is determined by the number of collagen fibres in our bodies (which is also genetic and so something else we can't change) and by our nervous system. There are numerous receptors in our muscles and tendons (we'll explore this later on in the book) that continually send sensory signals to the brain, providing information about how our muscles should move. If these mechanoreceptors detect that a joint may be unstable, they'll send a danger signal to the brain, and the brain will send a stop signal to the relevant muscles to stop this joint from moving. This prevents the joint from entering a potentially dangerous range of motion, protecting it from harm. Thus, some authors believe that stiffness (as a mechanical factor) is not a limiting factor of ROM and that mobility is instead limited by the nervous system (Piepoli, 2019).
- *Flexibility*: this is the ability of a material to bend without breaking (Nacleiro, 2011, within Peláez Maza, 2015a). By definition, improving flexibility is not the same as improving mobility, which in reality is what any athlete wants to do (although good mobility does require muscular flexibility to prevent injury).

So, do we really get 'tight' muscles? If tightness is a defence mechanism (against injury) and (static/passive) stretching attempts to turn off this mechanism, does it make any sense to stretch when that might increase our risk of injury? And will this help improve our ROM?

ACTIVE VS PASSIVE FLEXIBILITY AND MOBILITY RESERVE

The brain is like an overprotective parent who's obsessed with keeping us safe, but sometimes this excessive protection can have the opposite effect.

Luckily, the brain is constantly being fed information about the body (such as body position, tension, muscle length) and using this information to instruct us to move or not move, or perform a certain action or not. For example, if you sprain your ankle, the brain will not only issue pain signals to stop you from fully weighting your foot but will also orchestrate a new pattern of movement (limping) so you can still walk. All this information comes from numerous sensory receptors (proprioceptors) that are located around the body (in muscles, tendons, ligaments, joint capsules, and so on) and which are directly connected to the nervous system. To keep things simple, we'll just mention three different types:

- *Muscle spindles*: receptors which measure and report on variations in muscle length and the speed at which these variations occur.
- *Golgi tendon organs*: receptors found in tendons, next to the bone–tendon junction, which report on variations in muscle contractions.
- *Joint receptors*: receptors that are embedded in joint capsules, ligaments, and so on, which report on the position of the joint.

Using the information sent by these receptors, the brain controls the mobility of each and every joint in the human body. In practice, this range of motion is divided into:

- *Active flexibility*: the ROM we can achieve voluntarily. How far apart we can spread our legs or if we can bend over and touch our knees, our toes or get our hands flat on the floor. As muscle tension is determined by the brain based on the information received from mechanoreceptors, active ROM will vary depending on what action is being performed and when it is being performed (Latash and Zatsiorsky, 1993).
- *Passive flexibility*: the ROM we can achieve with the assistance of an external force that moves the joint into a certain position, such as a band, or our hands. However, there's a difference between the limit of our active ROM and the point at which the joint is at immediate risk of injury. This difference exists so when any sort of external force is applied (for example, a fall or a collision), the joint has a small margin of extra mobility before reaching breaking point.

The difference in mobility between active and passive ROM is referred to by some authors as our mobility reserve. It is the range of motion that we can't access voluntarily but which we could tap into with training.

MYOTATIC REFLEX AND AUTOGENIC INHIBITION/INVERSE MYOTATIC REFLEX

The body uses two basic reflexes, among other things, to control muscle tone:

- *Myotatic reflex*: this measures and controls muscle length. In response to stretching, this reflex causes the muscle to contract in order to resist the stretch.
- *Autogenic inhibition or inverse myotatic reflex*: this measures and controls muscle tension. In response to intense and prolonged tension (as created by passive stretching), the proprioceptors send signals to the brain to relax the muscle in order to prevent any damage. By relaxing the muscle, the brain inhibits its ability to produce force. As such, it doesn't seem that sensible to activate this reflex in climbing or in any other sport. In fact, inhibiting muscle response to improve joint mobility, precisely when we want the best possible response from our muscles (during sport or training), doesn't make any sense at all. This is why we should avoid

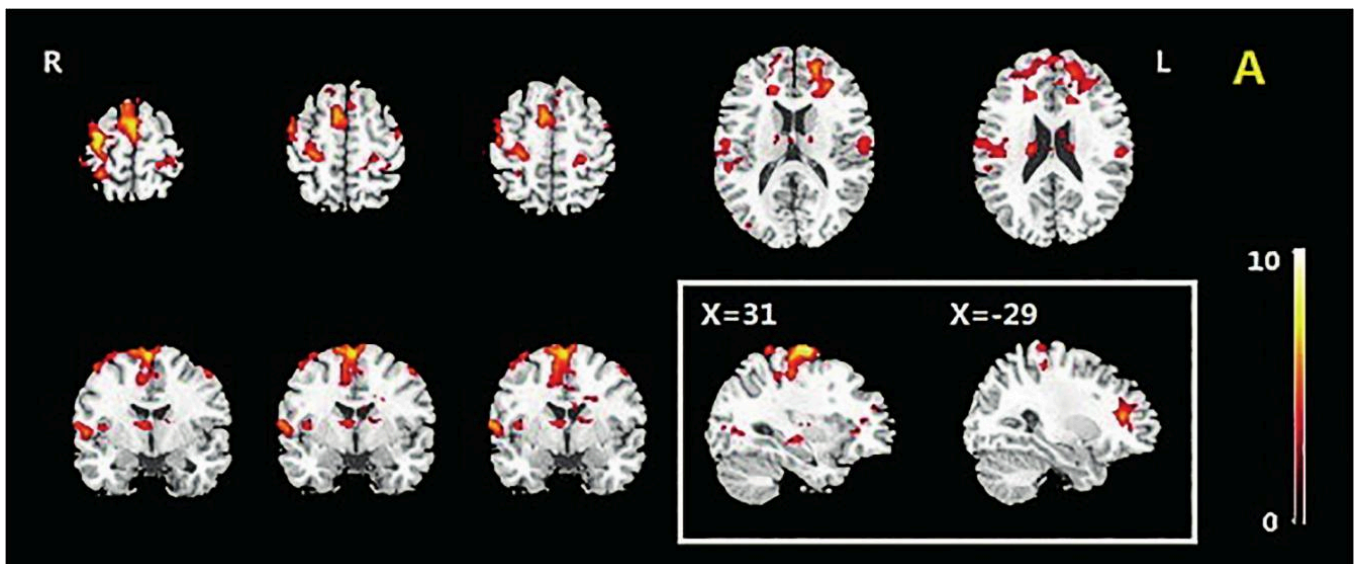
passive or static stretching of agonist muscles before or during training as it would send contradictory signals to the brain (contract, relax, contract, relax, and so on), which can easily lead to injury.

THREAT PERCEPTION AS A LIMITING FACTOR OF ROM

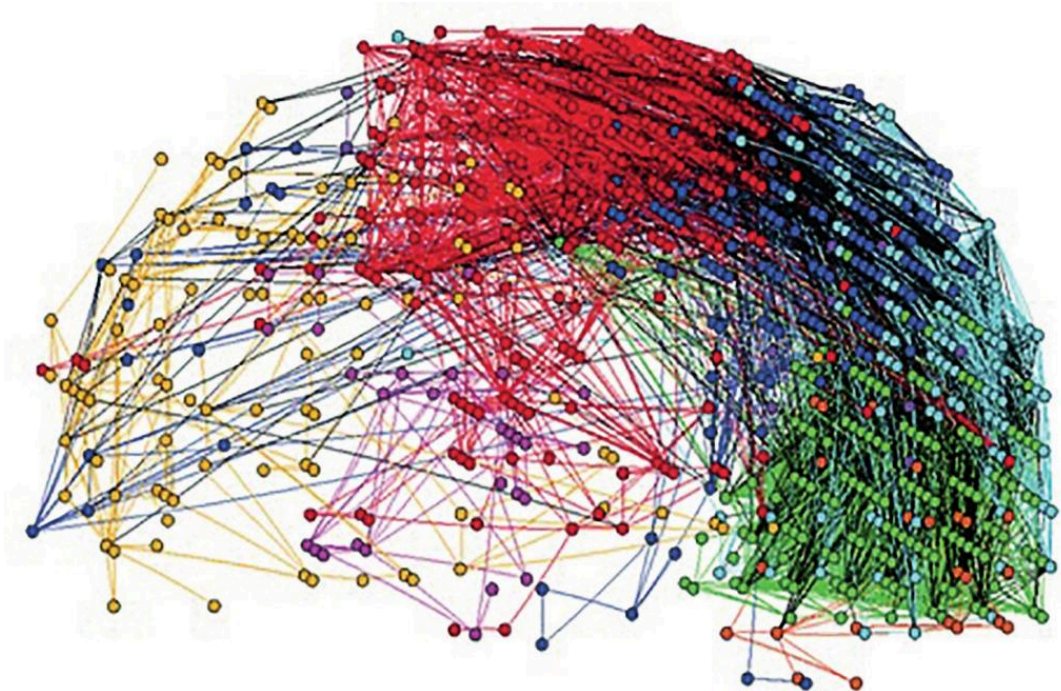
Why would the brain limit a joint's range of motion? The brain is an expert at making predictions. It anticipates all possible outcomes and will attempt to give the best possible response based on past experience. For example, if you're about to jump from a certain height, your brain will assess this height and produce what it thinks is the right amount of force to break your fall. This calculation is based on past experiences of you jumping from the same or similar heights. In response to stretching, mobilisation or the need to move, the brain considers all possible factors (such as how comfortable you are, if you're in a safe environment, the necessity of the action, good or bad past experiences) and forms a response. If, for whatever reason, it decides that moving a joint into a certain position means it will lose control of the joint, and if this lack of control could lead to injury, it will react in two ways: inhibiting the muscles responsible for moving the joint, and creating tension and pain in another place in order to bypass the 'zona incerta' in the brain, which it is proposed plays a role in gating sensory input and controlling pain.

BRAIN MAPS, SIMS AND DIMS

The brain is home to millions of interconnected neurons. It is the control centre for the whole body and decides how and when we move. To do this, it contains a representation of every muscle in the human body. When a group of neurons – a neural circuit – is activated, it controls a certain muscle. To perform a specific movement, several of these circuits are activated at the same time (to use a crimp above your head, you need to raise your arm, turn your hand, move your fingers, and so on). To perform a movement that you've done before, the brain uses a 'map' of these circuits to produce this movement.



ACTIVATION IN DIFFERENT AREAS OF THE BRAIN FOR FINGER FLEXION (LI ET AL., 2015)



REPRESENTATION OF BRAIN CONNECTIVITY (MEUNIER ET AL., 2010)

These maps are closely linked to the brain's powers of prediction as they are continually updated with new experiences. For example, when you throw for a hold on a new route, your brain activates its map for holding an edge. However, since you've never held this specific hold before, it doesn't know if this is the right map. Using the information it receives about visual perception, body position, and so on, it decides on what it thinks will be the best map for the job. When you make contact with the hold, your brain will readjust its map or create a new one, depending on the information received about the size, depth, texture and other characteristics of the hold.

But what's this got to do with mobility? It's simple: the brain will perceive a potential threat if it has to rely on a damaged or corrupted map of the movement that we want to perform (if an injury, awkward body movement or bad experience has damaged the map). A damaged map could be missing key information about the joint, in which case the brain will restrict its movement in order to prevent injury. Likewise, if there's no map for the movement we want to perform, this will also be seen as a potential threat and we'll need to inform the brain that the movement is safe.

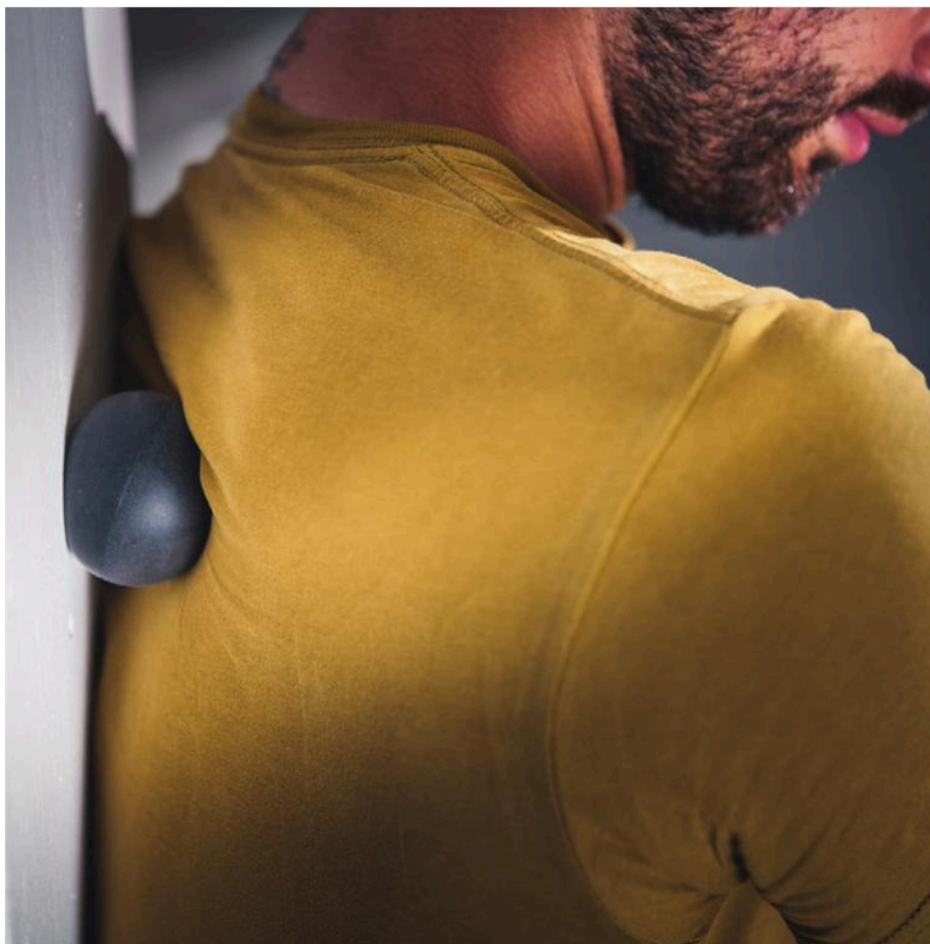
This is where things get interesting, with the appearance of SIMs and DIMs (Moseley and Butler, 2015). SIMs (Safety In Me) are things that make us feel strong, happy, more secure, and suchlike. DIMs (Danger In Me) are things that we feel might endanger our body, life, happiness, and so on. They function as positive or negative neurotags that are attached to actions or feelings and which activate different sets of brain maps. As such, they affect how the brain responds to different situations. As this is quite a complicated topic, let's use an example based on climbing. Helen was training at the climbing wall a few days ago. She did a drop-knee on a boulder problem, slipped and hurt her knee. She had some physiotherapy for the pain in her knee and she's now made a full recovery. She goes back to the wall and is about to try the same problem again. Her brain starts predicting what might happen: it knows that she hurt herself on this move before and, what's more, the brain map for this move is now damaged. This is when a DIM will appear, signalling danger: be on alert, watch out. She tenses up, her muscles become stiff and she can't move fluidly. She pulls on the problem and can't even get to the move to try the drop-knee. She needs to retrain her brain and create a SIM so that she isn't afraid of drop-knees and can move fluidly and confidently through them.

OPTIONS FOR OPTIMISING MOBILITY

After all this, what can we do to improve mobility? There are a number of options, and each works in a different way and is more or less useful depending on the person:

- *Breathing and relaxation techniques:* although it seems like the simplest thing in the world, breathing and relaxation can help improve joint mobility. Feeling stressed puts the sympathetic nervous system in a state of high alert, anticipating danger. As we saw earlier, if the brain senses danger, it won't let us do anything that it doesn't think is safe. What's more, on a cellular level, the shape and density of the fibroblasts in our fascia are affected by incoming electrical impulses (a phenomenon known as piezoelectricity), meaning they're affected by emotions and, of course, anxiety is a powerful emotion.
- *Techniques to reduce descending inhibition:* many physiotherapy techniques and paraphernalia such as foam rollers, massage sticks and different size/texture balls are based on the age-old principle of massage. This helps increase blood flow in the massaged area and therefore boosts oxygen supply (improving the predisposition of this area to exercise). It also stimulates the proprioceptors, creating a signal to activate the

mechanisms of descending pain inhibition (meaning less pain information reaches the brain). In turn, this can improve stretch tolerance and therefore ROM (Støve *et al.*, 2019).



- **Flossing:** this technique involves wrapping a stretchy band around a joint and moving the joint as much as possible for a set period of time (usually no more than two minutes). There are numerous benefits to this type of compression. It holds the joint firmly in place, so the brain knows that the joint is secure and under control. It also holds the superficial fascia in place and pulls on it when the joint moves. This causes the fascia to glide over the muscle, reducing adhesions and therefore improving both the transmission of force through the superficial and deep fascia system and the sliding of muscle fibres. Finally, flossing affects the blood supply to the targeted area. By moving the joint, we're increasing blood supply to the area, but by compressing it at the same time, we're stopping the blood from entering the area. (Warning: this is an effective technique but it can be highly uncomfortable, especially if the joint is injured in any way.) When the compression stops, this build-up of blood rushes through the area, carrying away metabolic waste and inflammatory substances, which also significantly reduces pain. This technique is also highly effective for recovering from injuries (Clements, 2015).



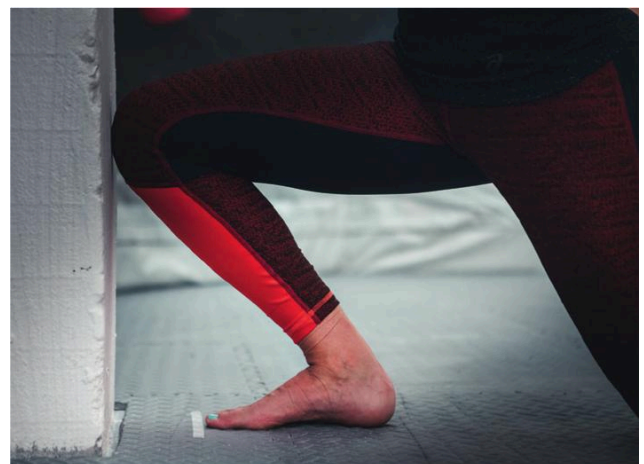
ASSESSING ANKLE MOBILITY (DORSIFLEXION).



WRAPPING THE JOINT FOR FLOSSING.



MOVING THE JOINT TO END RANGE OF MOTION IN ALL POSSIBLE DIRECTIONS.



MEASURING THE INCREASE IN ANKLE MOBILITY AFTER FLOSSING.

- **Improving end-range strength:** another example of the importance of strength. High-intensity isometric exercises (70 per cent of 1RM) have been proven to reduce intracortical inhibition in the contracted muscle, to reduce pain for up to 45 minutes after training and to increase corticospinal excitability, improving maximum voluntary contraction (Rio *et al.*, 2015, within Peláez Maza, 2016). In other words, these exercises give the brain control over the muscle, reducing the perception of threat. There is also strong evidence that submaximal eccentric exercises are highly effective for increasing muscle strength at a low risk of injury as they allow for greater motor control. (In fact, these types of exercises have been used for years in physiotherapy treatments.) As such, doing a few sets of submaximal eccentric contractions at the end range of movement will increase muscle strength and give the brain confidence and control in this range, therefore improving joint mobility. What is intracortical inhibition? As we already know, the brain can detect when a joint is in a potentially dangerous position and will send a pain signal and a signal that inhibits muscular force. The brain 'puts the brakes on' to stop the joint from moving towards the threat. To do this, it sends inhibition signals from the cerebral cortex. Through strength training, proprioception training and local vibration therapy, we can send a message to the brain to 'ease off the brake' and to 'give it some gas' instead, resulting in intracortical facilitation. This makes the area more sensitive, enhancing proprioceptive control, motor control and therefore strength, by reducing the effects of threat perception in the brain.
- **Local vibration:** most coaches won't have heard of this technique as it's more common in rehabilitation than in sports performance. There's an increasing amount of research into the benefits of this technique, although the

exact details of how it works remain unclear. A study performed by Souron *et al.* (2017) found that after eight weeks of local vibration therapy at 100Hz, on three non-consecutive days of the week, maximum strength increased in both the vibrated muscle and in the opposing muscle (vibration not applied). To explain these findings, it was suggested that the increase in signals sent by the proprioceptors to the brain led to greater activation of the corticospinal pathways. That is, it improved the interconnecting 'highway' between the brain and the muscles, allowing for more information to be sent and received, therefore improving motor control.





INITIAL ROM IN HIP ABDUCTION

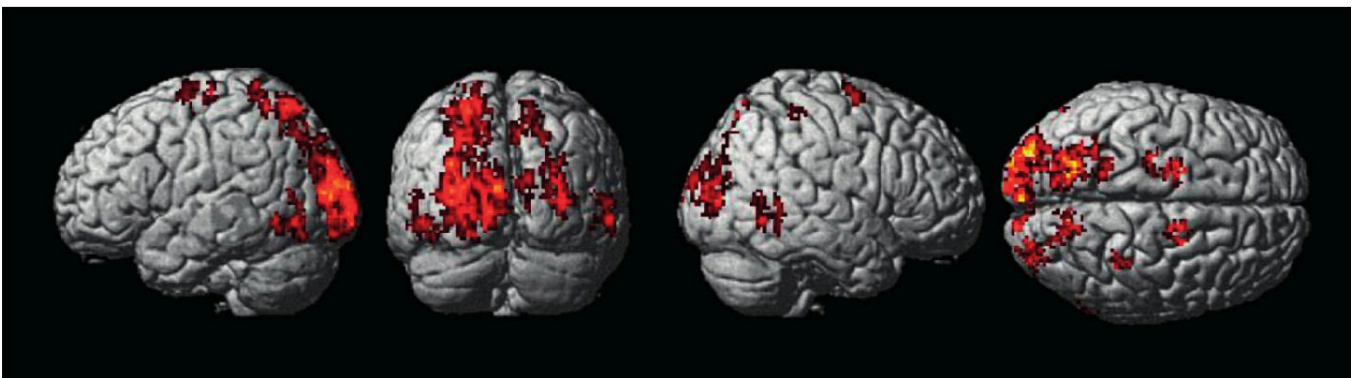


LOCAL VIBRATION + ISOMETRIC CONTRACTIONS IN ABDUCTOR MUSCLES



ROM IN HIP ABDUCTION POST INTERVENTION

- *Motor imagery*: recent studies have shown that in addition to mechanical factors, the nervous system and especially the brain also have a significant impact on ROM. Motor imagery (visualisation, imagery training, mental imagery, and suchlike) is defined as a cognitive task based on the internal reproduction, in this case mental, of a specific movement with no other type of motor stimulation (Cicinelli *et al.*, 2006, within Peláez Maza, 2015b). Athletes have been using these techniques for years to improve the quality of an action or movement and consolidate new techniques. In elite sport, visualisation is considered a fundamental part of training. But what's visualisation got to do with mobility? Although it might seem unlikely that you could improve your range of motion by simply imagining it, mental imagery does have an impact on brain maps. As such, it could be really useful in repairing damaged maps and in increasing cerebral control over our muscles.



CEREBRAL ACTIVATION OF HAND MOTOR MAPS DURING MOTOR IMAGERY EXERCISES (PILGRAMM *ET AL.*, 2016).

So where does this leave stretching? It's important to distinguish between the two main types of stretching: active stretching (resulting from our own actions – we stretch) and passive stretching (resulting from the application of external force – we get stretched).

The aim of passive stretching is to increase the length of the muscle by reducing muscular tension. But what really happens? In response to external force, the myotatic reflex shortens the muscle to protect it and stop it from tearing. And here's our first contradiction: you try to lengthen a muscle but end up shortening it. So how does this type of stretching work? It inhibits the muscle spindle, inhibits the myotatic reflex and activates the autogenic inhibition reflex. This means that the muscle can't respond, contract or produce any force. Although this can be useful in some situations, it definitely does more harm than good if you're about to do any type of sport.

As for active stretching, let's clear up a common misconception about the mechanics of stretching. The brain doesn't send a signal to stretch the targeted muscle, it sends a signal to shorten the opposing muscle. For example, if you sit on the floor with your legs straight and try to touch your toes, your brain won't send a signal to stretch your back and your hamstrings, but instead to contract the hip flexors. Therefore, the stretching of one muscle is an indirect consequence of the shortening of another.

Now let's distinguish between static stretching, and dynamic or ballistic stretching. Static stretching involves the isometric contraction of the opposing group of muscles to those which we want to stretch. There's different advice

on how long to hold these stretches, but a standard duration would be around 30 seconds. This is enough time to inhibit the myotatic reflex in the targeted muscle. However, as we've already seen, this isn't recommended if you're about to do any form of sport, except in very specific cases, and especially not for agonist muscles. The following studies show the effects of static stretching prior to physical exercise:

- Baxter *et al.* (2017) studied the effects of static stretching before running. The results showed that it decreased musculotendinous stiffness and elastic potential energy, reducing running economy and performance for up to an hour after stretching. It didn't reduce the duration or intensity of muscle stiffness or the prevalence of running-related injuries.
- Jelmini *et al.* (2018) performed a maximum strength test on the flexor muscles in both hands using a hand-held dynamometer. They then stretched the dominant hand only and repeated the test. The stretched hand lost 17.3 per cent of its strength. The non-stretched hand lost 10.8 per cent of its strength. This loss of strength continued for up to 15 minutes after stretching.
- Reduced performance after stretching has also been observed in other studies (Lauersen *et al.*, 2014; Kay and Blazevich, 2012; Thacker *et al.*, 2004). These studies found that stretching not only failed to reduce the risk of injury but could trigger an inflammatory response, which would damage soft tissue and therefore increase the risk of injury.

To conclude, static or passive stretching is simply not a good idea (either before or during training). Stretches that are held beyond the myotatic reflex (for longer than eight seconds) reduce the muscle's contractile capacity. They slow down the generation and transmission of electrical impulses. This can cause up to a 30 per cent loss in performance, which can continue for up to an hour after stretching and result in an increased risk of injury. So be very careful with this type of stretching.

Active stretches are repeated concentric contractions of the opposing muscles to those which we want to stretch. Instead of inhibiting the muscle spindle, these stretches activate the antagonist muscles. This generates intracortical facilitation, reduces inhibition and strengthens the muscles, while the brain feels safe and in control and therefore increases the range of motion: a much better scenario for performing at our best! Now we've covered mobility, let's turn our attention to how we can use movement to maximise the efficiency of our training.